Requirements Engineering & UML Use-Case Modelling

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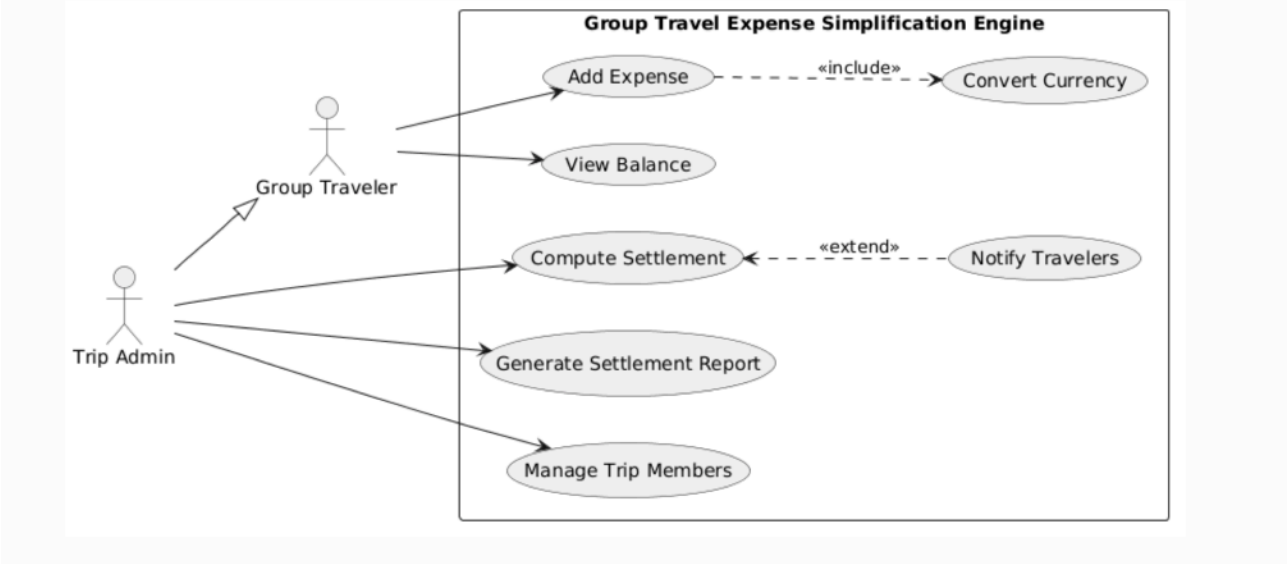
No. 37 - Group Travel Expense Simplification Engine

I. Functional & Non-functional requirements:

ID	Description	Priority	Acceptance Criteria	Rationale
FR-001	The system shall compute the optimal debt settlement graph using min-flow algorithms to minimize total peer-to-peer payment transactions.	High	Pass: Group balance settles to net zero with minimal payment edges. Fail: Sum of debts ≠ sum of credits.	Core value proposition — reduces the number of payments travelers must make to settle up.
FR-002	The system shall allow a Group Traveler to record an expense with an uneven split (custom amounts or percentages) among selected trip members.	High	Pass: Recorded expense's split amounts sum exactly to the total expense value. Fail: Split sum deviates from total by more than rounding tolerance (e.g. ₹0.01).	Real trip expenses are rarely split evenly; this is the primary data-entry use case.
FR-003	The system shall convert an expense entered in a foreign currency to the trip's base currency using the exchange rate for that expense's date.	High	Pass: Converted amount matches (original amount × published daily rate) within 2 decimal places. Fail: Conversion uses a stale or wrong-date rate.	Multi-currency handling is explicitly named in the problem context; incorrect conversion corrupts all downstream balance and settlement calculations.
FR-004	The system shall display each Group Traveler's real-time running balance (net amount owed or owed to them) for the trip.	Medium	Pass: Displayed balance updates immediately after any new expense or settlement is recorded. Fail: Balance is stale or inconsistent with the underlying expense ledger.	Travelers need visibility into their standing before the final settlement is computed.
FR-005	The Trip Admin shall be able to generate and export a settlement report listing all recommended payment transactions (payer, payee, amount, currency).	Medium	Pass: Exported report's transactions exactly match the min-flow settlement graph output. Fail: Report is missing transactions or contains transactions not in the computed graph.	Gives the group a shareable, actionable record to actually execute payments.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & security	The settlement calculation engine must execute in under 100 ms for groups of up to 50 travelers with 500 expenses, and all financial data must be encrypted in transit and at rest.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Settlement is computed frequently (on every new expense) and involves sensitive financial data, so both speed and protection are critical to usability and trust.
NFR-002	Usability & reliability	The system shall remain usable for balance viewing and expense entry when offline, queuing unsynced changes and reconciling them once connectivity is restored, without data loss.	Medium	Pass: Expenses entered offline appear correctly in the ledger after reconnection with no duplicates or losses. Fail: Any offline-entered expense is lost, duplicated, or silently dropped.	Group travel commonly happens in areas with poor connectivity (transit, remote destinations), so reliance on constant connectivity would break the core workflow.

II. UML Use-Case Diagram (PlantUML)



III. Use-Case Flow Specification

Use Case: Compute Settlement
Actor: Trip Admin
Preconditions: Trip exists with >= 2 members; at least one expense has been recorded; all expenses are converted to base currency.

Main Success Scenario:

1. Trip Admin selects "Compute Settlement" for the trip.

2. System retrieves all recorded expenses and current member balances.

3. System runs the min-flow settlement algorithm to compute the minimal set of payment edges.

4. System displays the settlement graph (payer → payee → amount) to the Admin.

5. Admin confirms the settlement.

6. System locks the settled expenses and updates all member balances to zero.

Postconditions: Settlement graph is generated and visible; member balances reflect net-zero state; settled expenses are marked immutable.

Alternate Flow — Currency Conversion Data Unavailable:
3a. If a live exchange rate is unavailable during recalculation, system falls back to the last cached daily rate for the affected expense and flags it as "estimated."
3b. Flow resumes at step 3 using the estimated value; the Admin is shown a warning banner listing flagged expenses.